

Differential Algebraic Equations

Suppose we have an ODE system
where some variables evolve much faster
than others:

$$y'(t) = f(y, z)$$

$$z'(t) = \mu g(y, z) \quad \mu \gg 1$$

Let $\varepsilon = 1/\mu$ and write

$$y'(t) = f(y, z) \quad (*)$$

$$\varepsilon z'(t) = g(y, z)$$

We call $(*)$ a "singular perturbation"
problem. If we consider the limit

$\varepsilon \rightarrow 0$, we get a system of ODEs and

algebraic equations:

$$y'(t) = f(y, z) \quad (**)$$

$$g(y, z) = 0$$

This is a differential-algebraic equation (DAE) system. They can appear in various forms; $(**)$ is called "semi-explicit" because we have separated the differential and algebraic parts.

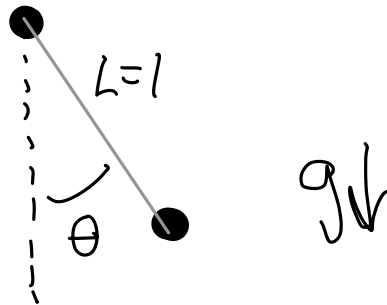
Examples from physical systems:

- Low mach number flow in the incompressible limit
- Fluid dynamics with very fast chemical reactions

Working with the DAE directly can be advantageous if we don't care about capturing the fast timescale.

The Pendulum

Algebraic constraints may also arise directly. Consider the rigid pendulum:



From Newton's laws we can derive the ODE:

$$\Theta''(t) = -g \sin(\Theta(t))$$

But if we want to stick to Eulerian coordinates we could instead write:

$$X'' = -Z(x,y)X$$

$$Y'' = -Z(x,y)Y - g$$

$$X^2 + Y^2 = 1$$

Coord. system centered at top of pendulum.

Here $z(x,y)$ is some function to be determined so that the constraint is satisfied.

Writing this as a 1st-order system we have

$$x' = u \quad y' = v$$

$$u' = -zx \quad v' = -zy - g$$

$$x^2 + y^2 = 1.$$

Let's differentiate the constraint:

$$2xx' + 2yy' = 0$$

$$\Rightarrow xu + yv = 0 \Rightarrow \vec{x} \cdot \vec{u} = 0$$

so motion is tangent to a circle.

Differentiate again:

$$x'u + xu' + y'v + yv' = 0$$

$$u^2 + v^2 - z(\underbrace{x^2 + y^2}_{=1}) - gy = 0$$

$$Z = u^2 + v^2 - gy$$

Kinetic - Potential
 Z is the Lagrangian!

Differentiate again:

$$\begin{aligned} Z'(t) &= Zu u' + Zv v' - gy' \\ &= -Zu z_x - Zv z_y - Zvg - vg \\ &= -Zz(u x + v y) - 3vg \end{aligned}$$

$$Z'(t) = -3vg$$

So we have the equivalent ODE system:

$$\begin{aligned} x' &= u & y' &= v \\ u' &= -Zx & v' &= -Zy - g \end{aligned}$$

$$Z'(t) = -3vg$$

We had to differentiate the constraint 3 times, so we call this a DAE of index 3.

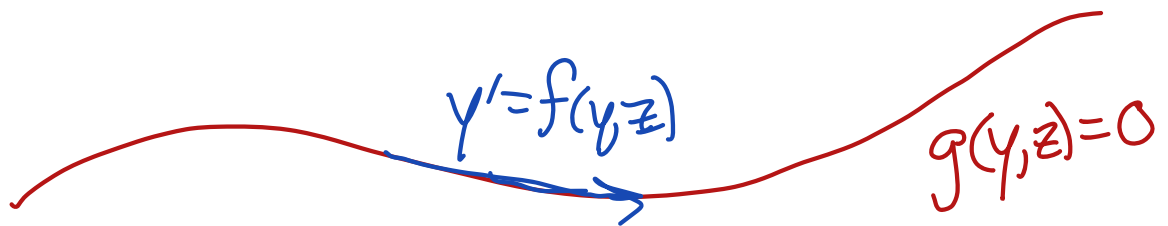
The index of a DAE strongly affects the choice of numerical method to solve it.

A DAE

$$y' = f(y, z)$$

$$g(y, z) = 0$$

with $y \in \mathbb{R}^m$, $z \in \mathbb{R}^n$, describes a system in $m+n$ dimensions that evolves on a m -dimensional manifold



An implicit DAE takes the form

$$F(w, w'(t)) = 0$$

where F takes values in $\mathbb{R}^n$ while $w \in \mathbb{R}^{m+n}$

Introducing $z = w'$ gives a semi-explicit form:

$$w'(t) = z(t)$$

$$F(w, z) = 0.$$

So when we discuss the index we need to specify the form. Semi-explicit index-2 is equivalent to implicit index 1.

The standard numbering is based on semi-explicit form.

Another example

$$u'_1 = u_3$$

$$(1-u_2)u_2=0 \Rightarrow u_2=0 \text{ or } u_2=1$$

$$u_1 u_2 + (1-u_2)u_3 - t = 0$$

If $u_2=0$: $u_3=t \Rightarrow u'_3=1 \Rightarrow$ index 1 (semi-explicit)

If $u_2=1$: $u_1=t \Rightarrow u'_1=1 \Rightarrow u''_1=0$ index 2
 $u'_1=u_3 \Rightarrow u'_3=u''_1=0$

General Rules for the index

Semi-explicit! $y'(t) = f(y, z)$

$$g(y, z) = 0$$

Has index 1 iff g_z is non-singular,

since:

$$g_y y' + g_z z' = 0$$

$$\Rightarrow z' = -g_z^{-1} g_y f$$

What if g is independent of z ? $g = g(y)$

$$g_y f = 0$$

Differentiate:

$$\Rightarrow g_{yy}f + \cancel{g_{yz}z'} + g_y f_y f + g_y f_z z' = 0$$

We get index z if $g_y f_z$ is nonsingular.

Incompressible fluid dynamics

In fluids, we have two important timescales:

- Fluid motion $\sim \frac{L}{u}$
- Acoustic (pressure) waves: $\sim \frac{L}{c}$

Often, $u \ll c$ (low Mach number flow), so acoustic time scales are very fast. If we imagine that acoustic waves are infinitely fast, we get the incompressible approximation. In 2D:

$$u_t + uu_x + vv_y + p_x = \nu \nabla^2 u$$

$$v_t + uv_x + vv_y + p_y = \nu \nabla^2 v$$

$$u_x + v_y = 0 \quad \text{Div-free constraint}$$

We have evolution equations for u, v but not p .

Semi-discretization: $W = \begin{bmatrix} U \\ v \end{bmatrix}$

$$W'(t) = N(W) - GP + vLW$$

$$DW = 0$$

Algebraic variables: P

Constraint doesn't depend on P , so this can't be index-1. It's index-2 as long as DG is non-singular.